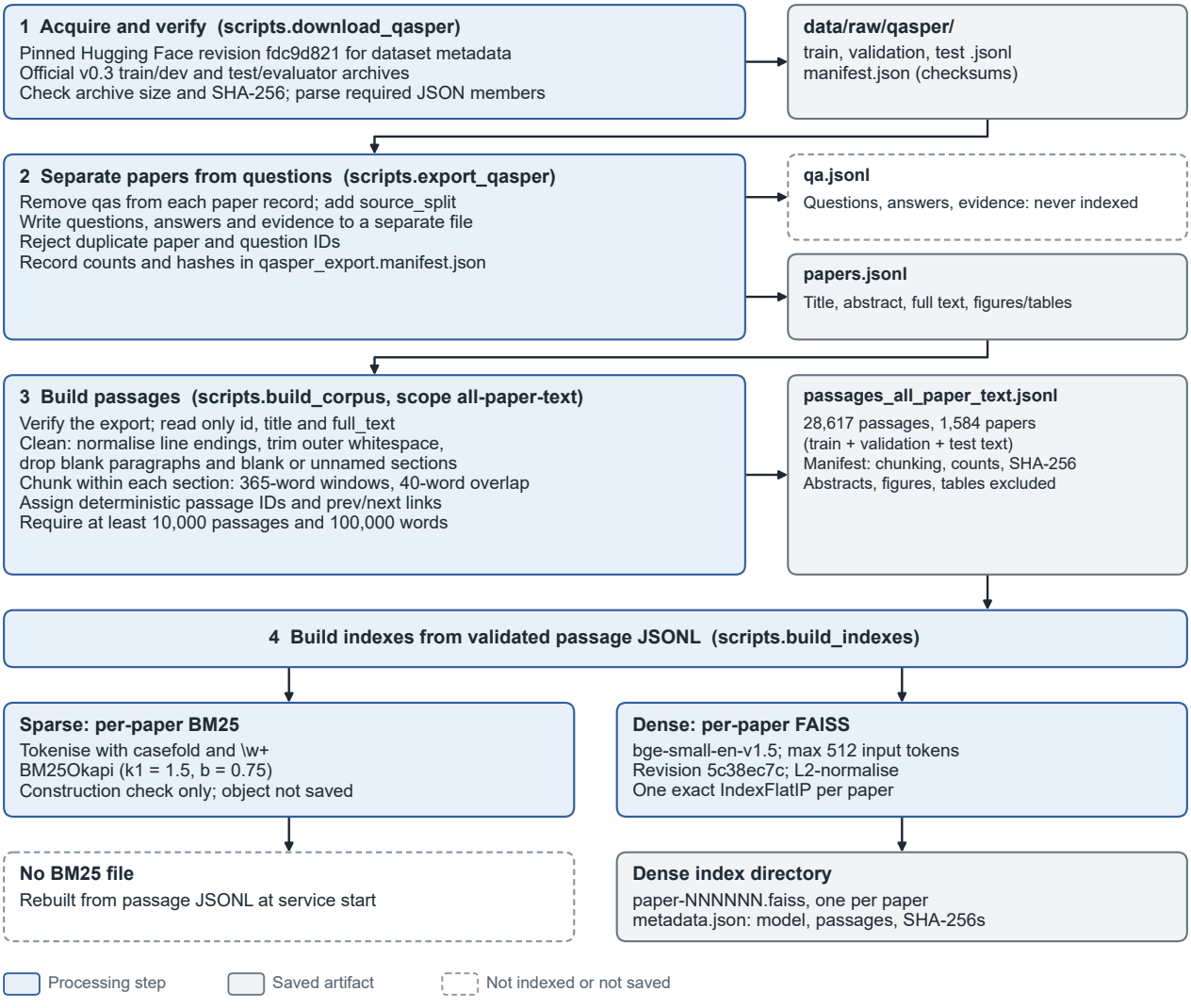

1 Acquire and verify (scripts.download_qasper)
Pinned Hugging Face revision fdc9d821 for dataset metadata
Official v0.3 train/dev and test/evaluator archives
Check archive size and SHA-256; parse required JSON members
data/raw/qasper/
train, validation, test .jsonl
manifest.json (checksums)
2 Separate papers from questions (scripts.export_qasper)
Remove qas from each paper record; add source_split
Write questions, answers and evidence to a separate file
Reject duplicate paper and question IDs
Record counts and hashes in qasper_export.manifest.json
qa.jsonl
Questions, answers, evidence: never indexed
papers.jsonl
Title, abstract, full text, figures/tables
3 Build passages (scripts.build_corpus, scope all-paper-text)
Verify the export; read only id, title and full_text
Clean: normalise line endings, trim outer whitespace,
drop blank paragraphs and blank or unnamed sections
Chunk within each section: 365-word windows, 40-word overlap
Assign deterministic passage IDs and prev/next links
Require at least 10,000 passages and 100,000 words
passages_all_paper_text.jsonl
28,617 passages, 1,584 papers
(train + validation + test text)
Manifest: chunking, counts, SHA-256
Abstracts, figures, tables excluded
4 Build indexes from validated passage JSONL (scripts.build_indexes)
Sparse: per-paper BM25
Tokenise with casefold and \w+
BM25Okapi (k1 = 1.5, b = 0.75)
Construction check only; object not saved
Dense: per-paper FAISS
bge-small-en-v1.5; max 512 input tokens
Revision 5c38ec7c; L2-normalise
One exact IndexFlatIP per paper
No BM25 file
Rebuilt from passage JSONL at service start
Dense index directory
paper-NNNNNN.faiss, one per paper
metadata.json: model, passages, SHA-256s
Processing step
Saved artifact
Not indexed or not saved